The real impact of AI Isn't in the Technology. It's in your business processes.

How to create value using AI in business processes

Koen Klokgieters | InnoPlot

Introduction

Most conversations about AI are focused on the wrong thing.

We debate models. We compare benchmarks. We follow the announcements from the hyperscalers with a mix of excitement and anxiety. And in doing so, we create the impression that the impact of AI is primarily a technology question something for the CTO's office to sort out.

It isn't.

The true impact of AI becomes visible only when you look at the activities that make up your organisation. Not the systems you run, not the platforms you've licensed but the *activities*. The things your people and your processes actually *do* every day to create value.

It is not technology itself that transforms organisations. It is the shift of tasks from human to machine and knowing where that boundary will settle.

A Framework that has survived every technology Wave

Michael Porter gave us the concept of the value chain: the idea that organisations are, at their core, systems of activities. Not products. Not org charts. Activities.

When you look carefully at those activities across any industry, any function, any organisation, you find the same seven building blocks as a foundation for the activity system of an organisation:

1. *Sensing* — reading the environment. Picking up signals, data, and information from inside and outside the organisation.
2. *Analysing* — making sense of what you've sensed. Recognising patterns, correlations, and anomalies. Converting raw data into insight.

3. *Deciding* — making choices. Weighing values, priorities, risks, and competing interests to arrive at a course of action.
4. *Executing* — making it real. The physical or digital realisation of a decision: a product made, a service delivered, a transaction processed.
5. *Learning* — adapting. Adjusting knowledge, behaviour, processes, and strategy on the basis of what the execution revealed.
6. *Communicating* — connecting. Transferring information, instructions, and expectations between people, machines, systems, and organisations.
7. *Coordinating* — aligning. Synchronising activities, people, and resources across time and space to achieve a shared goal.

These seven activities are the operating system of every organisation. And they are the lens through which the impact of AI becomes legible.

Every technology revolution has shifted the same boundary

Here is the pattern history shows us and it is remarkably consistent.

Every major technological revolution moves a larger share of the activity system from human hands to machines and systems.

The industrial revolution mechanised *executing* and parts of *sensing*. Machines took over physical labour. Humans retained everything cognitive: analysing, deciding, learning, communicating, coordinating.

The computing revolution extended automation into *analysing* and routine *deciding*. Computers began to record, calculate, and manage processes within fixed parameters. Humans shifted their energy to exceptions and interpretation, the things the rules couldn't handle.

The internet revolution brought *communicating* and *coordinating* into the machine's domain. Digital platforms handled information exchange and alignment in real time. Human coordination moved up the value chain, towards relationship management and the governance of exceptions.

Each wave took something that felt irreducibly human and made it automatable. Each wave left a residual human role that felt at least temporarily safe.

AI is the current wave. And it is different.

Why the AI-technology Is unlike the others

Previous technologies processed and connected information. AI can *reason* about it.

That is not an incremental improvement. It is a qualitative shift.

The core capabilities that make this possible:

- The foundation is ***machine learning*** (ML). This means that computers learn from large volumes of data. They independently search for patterns and continuously improve through practice and error correction.
- ***Neural networks*** function somewhat like the human brain. They consist of layers of “neurons” that adjust their settings during the learning process.

- ***Deep learning*** uses many of these layers stacked on top of one another, enabling computers to perform complex tasks such as image recognition and language understanding.
- ***Generative AI*** continuously predicts the most likely next piece of information (for example, the next word in a sentence). By repeatedly making such predictions, it is able to generate new text, images, or other content.
- ***Autonomous AI / AI Agents*** execute tasks independently and make decisions in chains.
- ***Computer Vision AI*** enables image recognition and visual inspection based on deep learning.
- ***Robotics & RPA*** (Robotic Process Automation) concerns the software-based or physical automation of structured and repetitive tasks without human involvement.
- ***Reasoning Models / Neurosymbolic AI*** combine statistical AI with logical reasoning and are particularly relevant for complex decision-making tasks.

Three enabling conditions made all of this possible: exponentially growing computing power (Moore's Law, then GPUs and TPUs); the explosion of available data from the internet, smartphones, and sensors; and algorithmic breakthroughs such as backpropagation (a method for correcting errors during training) and transformer architectures that made training large models practical.

The result is that AI is now beginning to automate *learning*, *analysing*, and increasingly *deciding*, the cognitive activities that previous waves left to humans.

How AI Is transforming each of the foundation activities

Let's be concrete. Here is where AI is already operating and where it is heading across each of the seven activity types.

1. Sensing

AI excels at sensing when signals are digitalisable, volumes are high, objectivity matters, and measurement needs to be continuous.

In practice: visual quality inspection in factories, anomaly detection in sensor data, real-time monitoring of network traffic, satellite imagery analysis. These applications are not experimental. They are technically mature and are being deployed at scale today.

2. Analysing

AI is most effective in analysis when there is sufficient historical data, when patterns are statistically stable, and when the number of variables exceeds human cognitive bandwidth.

In practice: demand forecasting, credit risk modelling, fraud detection, predictive maintenance, real-time analysis of customer behaviour. Generative AI is increasingly used to summarise large volumes of reports and extract insight from unstructured text, a capability that is rapidly maturing.

3. Deciding

AI can own decision-making when criteria are explicit and learnable, when the normative or legal stakes are limited, and when decisions are frequent and outcomes measurable.

In practice: credit scoring, dynamic pricing engines, inventory replenishment, production scheduling, content recommendation. The boundary here is critical: AI handles high-frequency, rules-based decisions

with increasing confidence. Infrequent, high-stakes, normatively complex decisions remain human for now.

4. Executing

This is where AI's advantage is most immediately visible. Repetitive, structured, digital execution is already highly automatable.

In practice: data processing and transformation, transaction handling, document routing via RPA, robotic assembly lines, autonomous logistics vehicles. Generative AI adds a new dimension: the automatic drafting of reports, correspondence, and even production-grade software code.

5. Communicating

AI handles communication well when it follows predictable formats, when scale and availability matter more than depth, and when trust relationships are not the primary requirement.

In practice: customer service chatbots, automated reporting, real-time translation, personalised outreach at scale. The risk here is conflating efficiency with effectiveness , AI can produce volume and speed, but relationship-rich communication still requires human judgement and emotional intelligence.

6. Coordinating

AI is powerful in coordination when the challenge can be framed as an optimisation problem and when the relevant data is digitally available.

In practice: supply chain optimisation, last-mile logistics routing, shift scheduling, capacity planning. As organisations become more data-rich, the surface area for AI-driven coordination expands significantly.

7. Learning

Perhaps the most profound impact of all. AI can learn at a rate, scale, and consistency that no human organisation can match as long as the feedback loops are measurable and the objectives are defined.

In practice: self-improving recommendation engines, A/B testing at machine speed, production systems that continuously retrain on new data. The critical constraint: AI learns *towards a defined objective*. Setting that objective, and determining when it is the right one, remains a human responsibility.

Overview

Activity	Rule-based	Machine Learning	Generative AI	Autonomous Agents	Computer Vision	RPA/Robotics	Reasoning
Sensing	●●●	●●●●	○	●●	●●●●●	●●	○
Analysing	●●	●●●●●	●●●	●●	●●●	○	●●●
Deciding	●●●	●●●	●●	●●	○	○	●●●●
Executing	●●	●●	●●●	●●●	●●●	●●●●●	○
Communicating	●●●	●●●	●●●●●	●●●	○	●●	●
Coordinating	●●●	●●●●	●●	●●●	○	●●●	●●
Learning	●	●●●●●	●●●	●●●	●	○	●●●

●●●●● = primary application / highest relevance → ○ = not or barely applicable

Where the boundary sits: Primary, Supporting, and Management Processes

Not all processes are equally exposed. The degree to which AI can replace human activity depends fundamentally on one question: *how clearly can the objective of the activity be defined?*

Primary processes — those that directly create value for the customer have relatively clear, measurable outcomes. A product either meets specifications, or it doesn't. An order is either delivered on time, or it isn't. The feedback loop is short and objective. This makes primary processes highly amenable to AI. The combination of Computer Vision, RPA, Machine Learning, and generative AI means that most executing and sensing activities here can already be replaced or will be within the near term. The highest replaceability in the entire framework sits here.

Supporting processes — HR, finance, IT, procurement are highly standardised and data-rich, making them attractive targets for automation. Machine learning and RPA dominate the executing core of these functions. The difference from primary processes is that supporting processes involve more human coordination at the edges: HR advisory conversations, legal judgement calls, procurement negotiations that require relationships as much as logic. Replaceability is high in the centre, lower at the periphery.

Management and control processes — governance, risk, audit, compliance are fundamentally different in character. Here, the challenge is not executing a defined task but *forming a judgement about direction and integrity*. Risk management requires identifying risks that haven't been defined yet. Governance demands accountability to stakeholders with conflicting interests. Audit requires professional scepticism and independence. AI can support all of these with data, signals, and pattern recognition. But the normative judgement, the step that carries accountability, remains irreducibly human.

Dimension	Primary processes	Supporting processes	Management & control processes
Core activity	Creating value for the customer	Facilitating the organisation	Safeguarding direction and integrity
Objective definition	Clear and measurable	Largely standardised	Normative and context-dependent
Dominant AI role	Replace & execute	Automate & support	Signal & advise
Strongest AI types	RPA, Computer Vision, ML	ML, RPA, Generative AI	ML (analysis), Reasoning (emerging)
Weakest AI types	Reasoning, autonomous agents	Autonomous agents	Generative AI (risk: false certainty)
Replaceability of execution	Very high	High	Low to moderate
Replaceability of judgement	Moderate	Low	Very low
Human role	Directing, managing exceptions, ethics	Advisory, relational, norm-setting	Accountability, judgement, governance
Primary AI risk	Quality loss in unexpected situations	Bias in HR/legal AI	False objectivity in normative decisions

The next five years: Three trends that will reshape every organisation

The shift underway is not linear. Three trends are converging simultaneously, each accelerating the others.

Trend 1: From AI Assistant to AI Colleague

The rise of **agentic AI** systems that don't just respond to prompts but plan and execute multi-step workflows autonomously represents the most disruptive near-term development.

Today's generative AI writes text, answers questions, and produces analysis on demand. Tomorrow's AI agents will *complete workflows*. An agent receives a customer enquiry, gathers the relevant information, drafts a proposal, processes the order, and monitors progress without a human touching the process.

In HR, agents will assess applications, schedule interviews, generate onboarding documentation, and coordinate the process end-to-end. In finance, agents will detect anomalies, prepare reconciliations, and draft management accounts.

The shift is not from person to software. It is from person to *digital colleague*.

The question is fundamental: *where does human-in-control remain?* And who is accountable when the agent makes a mistake?

Trend 2: The Real-Time Organisation

Organisations have traditionally operated on the rhythm of reports: monthly accounts, quarterly reviews, annual strategies. That rhythm was set by the cost and latency of information processing.

AI removes both constraints. **Processes can be monitored continuously. Adjustments can be made in real time.** Risks can be surfaced before they materialise.

In primary processes: dynamic pricing, autonomous inventory management, real-time production scheduling. In supporting processes: financial anomaly detection that doesn't wait for month-end, HR systems that predict attrition before it happens, IT platforms that prevent outages before users notice them. In management processes: risk signals that emerge from pattern recognition rather than backward-looking reports.

The organisation moves from *reactive to predictive*. The new challenge is not data or processing power it is governance. *How do you embed ethical and moral values into a system making thousands of autonomous adjustments per day?*

Trend 3: Hyper-Personalisation at Scale

The third trend is the collapse of the trade-off between personalisation and scale. AI makes it possible to tailor products, services, communication, and decisions to the individual at volumes that were previously impossible.

For customers: personalised services, individual pricing, bespoke production runs, dynamic interactions that adapt in real time. For employees: personalised learning pathways, tailored support, work processes calibrated to individual role and performance. For management: an increasing proportion of decisions that are made differently for different individuals, situations, and contexts.

This creates a governance challenge that is growing at the same rate as the capability. As the number of individual decisions expands, so does the need for transparency and explainability.

The question is shifting. It is no longer *"Can AI take this decision?"* It is *"Can we explain why this decision was taken?"*

The Dark Factory: a working prototype of what Is coming

The **dark factory** is a manufacturing facility that runs lights-out, no human operators on the floor, no manual handling, no shift breaks. Robotic arms assemble components. Autonomous vehicles move materials. Machine-vision systems inspect quality. All of it coordinated by integrated software in real time.

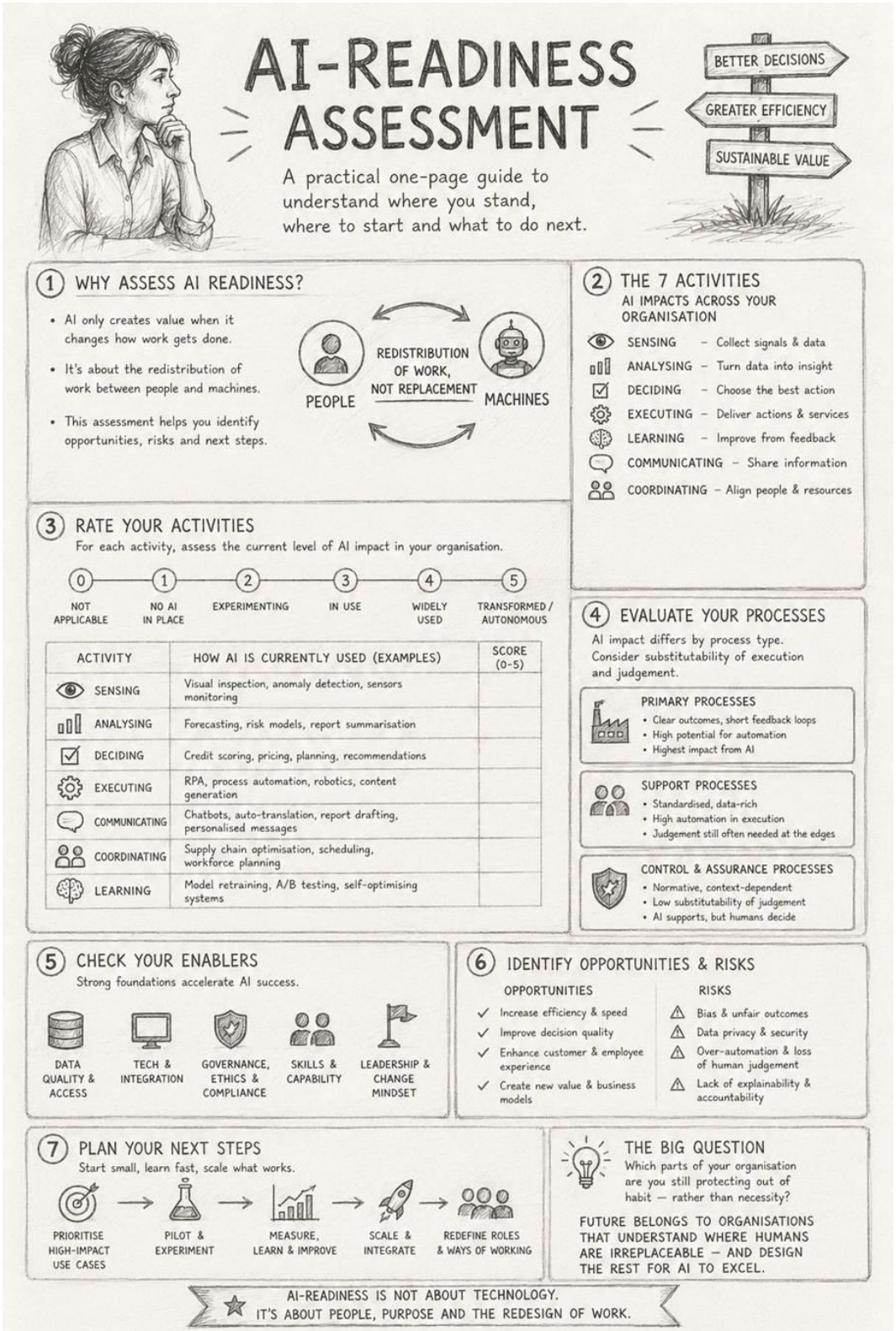
Until recently, this required a specialist AI system for each individual function: one for defect detection, another for scheduling, a third for maintenance prediction. Each operated in isolation. The current generation of AI removes those boundaries.

In a dark factory today, a single agentic workflow can detect a quality anomaly, trace its root cause in process data, adjust machine parameters, reschedule the affected batch, update the digital twin, generate a maintenance instruction, and notify the supply chain in seconds, with no human in the loop.

This is not a glimpse of a distant future. It is operational today in leading manufacturers. And it is the template for what will arrive, at different speeds, in every sector.

The dark factory is not the future of manufacturing. It is a working prototype of the AI-orchestrated organisation.

AI-readiness scan for your organization



The Conclusion: That most leaders are not ready

Here is the paradox at the heart of this transition.

As machines become better at analysing and executing as they become more capable of handling complexity, scale, and speed the human role does not diminish. It becomes more important. But it concentrates in a different place.

The activities that will remain irreducibly human are precisely those where the objective function itself must be defined: where values, interests, accountability, and legitimacy are at stake. Not because humans are better at these things than machines. But because they are the activities where *being human* with all its partiality, judgement, and accountability is the point.

The organisations that will succeed over the next decade will not be those with the most AI. They will be those with the clearest understanding of which activities to automate, which to protect, and how to make the two reinforce each other.

The future does not belong to the organisations with the most sophisticated AI.

The future belongs to the organisations that understand where human intuition, knowledge, skills, and experience remain irreplaceable.

About the Author

Koen Klokgieters is the founder of InnoPlot and has spent more than three decades working at the intersection of strategy, innovation, technology and organisational transformation. His experience combines corporate leadership, consulting, entrepreneurship and education.

Through InnoPlot, he supports organisations and their leaders in translating strategic challenges into innovation capabilities, organisational change and measurable impact.

InnoPlot | Business & Technology Innovation (innoplot.com)

Note: Full references are provided in the complete academic version of this paper. The core ideas, analysis, and narrative are the author's own, while generative AI tools such as ChatGPT, Claude, and Perplexity were used to support the writing and research process.